

Shadow Futures: Contribution Uncertainty and the Self-Reinforcing Market

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First posted December 2025; revised October 2026

Abstract

Can a market observe productive inputs perfectly yet fail to learn their effects on reward? This paper studies adaptive allocation in which verified inputs enter reward odds while past rewards change future exposure. The comparison budget is cumulative allocation probability outside the current leader. In conditional logit, it bounds Fisher information and finite-horizon statistical separation. Under a common predictable design, local equivalence, and comparison-dominated separation, an almost-surely finite budget yields equivalent complete-history laws. No estimator of a nonconstant contribution functional can then be consistent at every parameter; no test can have both errors vanish. By Blackwell–Dubins, conditional forecasts nevertheless merge: predictive agreement can coexist with permanent uncertainty about input effects. A local two-point lower bound relates attribution precision to the expected comparison budget. Strong reinforcement supplies one sufficient case; fixed inputs with latent position yield a separate failure of point identification. Unrealized comparison paths are shadow futures. The contribution is an economic information bound and its attribution interpretation, using classical probability machinery. Productive inputs can remain causally effective even when their effects cannot be consistently recovered from one market history.

Keywords: comparison budget; increasing returns; path dependence; identification; cumulative advantage; monopoly; competition policy

JEL classifications: C13; C18; D43; D83; D85; L41

1. Introduction

A successful firm can arrive with a warehouse of evidence. It can produce payroll records, patents, code commits, invoices, and proof of capital committed and risks borne. Every document can be genuine. The missing exhibit is the same firm run again with a different first customer, ranking, distribution channel, or early shock. That firm exists only as an unrealized history.

Economic systems often turn early outcomes into later access. A freelancer's first rating affects the next client; a creator's first burst of attention affects recommendations. A startup's first customer can shape financing and distribution. A paper's first placement can shape later citations. Field experiments document success-breeds-success effects (van de Rijt et al., 2014). Real work and risk occur inside tournaments in which each result changes later odds.

Economics has long understood that increasing returns and cumulative advantage can create concentration, lock-in, and historical selection (Arthur, 1989; David, 1985; Merton, 1968). This paper asks a prior question. Once reward compounds through position, can the realized market history reveal how much of the outcome was caused by work, quality, capital, or risk?

The distinction is between causal effect and recoverability. Better products may raise winning odds; effort and capital at risk may improve outcomes and justify ex ante premiums. These effects need not be consistently recoverable from one reinforced history. Abundant detail need not reveal intervention effects (Bar-Yam, 2016). A market can keep paying after it has stopped learning.

Akerlof's (1970) lemons problem concerns hidden quality. Here productive inputs may be public, measured without error, and used directly by the allocation rule. The missing information

is historical: how those inputs would have performed under different early shocks. Better disclosure cannot reveal an experiment that was never run.

I call this contribution uncertainty: uncertainty about a model-defined effect of productive inputs on reward. A time sheet, audit, or ledger may establish that work occurred. It does not establish the counterfactual difference that work made through a self-reinforcing allocation process.

The paper makes three claims. First, transaction volume need not measure information about input effects. In conditional logit, cumulative Fisher information and finite-horizon statistical separation are bounded by the comparison budget. Second, under the stated equivalence and separation conditions, an almost-surely finite budget rules out universally consistent recovery of nonconstant contribution functionals from one history. Third, competitive routes can preserve informative comparison when they share a structural parameter and generate independent allocation shocks. Concentration can therefore create an epistemic loss without deliberate exclusion or monopoly pricing.

The probability machinery is classical, from product-measure equivalence (Kakutani, 1948) to general random sequences (Kabanov, Liptser, and Shiryaev, 1977). Reinforced-process theory explains nonlinear dominance (Pemantle, 2007); Le Goff and Soulier (2017) already establish single-trajectory estimation failure for a reinforced urn. Blackwell and Dubins (1962) establish predictive merging under absolute continuity, and predictive agreement with estimation failure appears in geostatistics (Zhang, 2004). The contribution here is the comparison-budget bound for adaptive allocation with observed productive inputs, its finite-horizon precision limit, and its economic interpretation.

2. The Self-Reinforcing Market and Its Comparison Budget

There are n alternatives. Before reward $t + 1$ is allocated, alternative i has a verified input profile x_{it} . The profile may include output, hours, costly effort, measurable quality, capital committed, foregone income, or exposure to loss. The profiles may change with the observed history. The contribution parameter β gives the direct loading of those inputs on reward odds. A predictable state index s_{it} records accumulated position, including prior sales, reputation, citations, financing, ranking, or platform placement.

For every β , the experiment is defined on the same raw filtered space, with terminal sigma-field $F_\infty = \sigma(\cup_t F_t)$. The initial law is common, and the input profiles and state indices are common predictable functions of the realized allocation history. All parameter-dependent randomness in the next observed allocation enters through the recipient kernel below. If additional observed variables have parameter-dependent laws, their information must be included separately.

$$\Pr_\beta(J_{t+1} = i \mid \mathcal{F}_t) = p_{it}(\beta) = \frac{\exp(x_{it}^\top \beta + s_{it})}{\sum_{j=1}^n \exp(x_{jt}^\top \beta + s_{jt})}. \quad (1)$$

Verified input differences enter relative reward odds directly. The state term permits past reward to change future access.

At date t , let the current leader be any alternative with the largest conditional allocation probability. Define residual contestability as the probability that the next reward goes elsewhere:

$$\varepsilon_t(\beta) = 1 - p_{w_t t}(\beta) = 1 - \max_i p_{it}(\beta). \quad (2)$$

I call the cumulative residual contestability through date T the comparison budget:

$$B_T(\beta) = \sum_{t=0}^{T-1} \varepsilon_t(\beta), \quad B_\infty(\beta) = \sum_{t=0}^{\infty} \varepsilon_t(\beta). \quad (3)$$

T counts transactions; B(T) sums residual contestability. A constant leader probability of 0.999 adds one unit per thousand transactions but gives an infinite budget over an infinite horizon. Concentration alone does not establish finite comparison, and an infinite budget does not guarantee identification. Empirical use requires independently justified estimates or bounds on allocation probabilities; market shares alone do not reveal them. Any additional information used to obtain such bounds belongs in the observation model.

The estimand is a nonconstant functional $\psi(\beta)$, such as the change in expected cumulative reward under a specified input intervention and horizon. Its causal interpretation requires the structural allocation rule and feedback dynamics to remain valid under that intervention. Realized profit, market share, or audience size is one draw from this system, not the functional itself.

3. The Comparison-Budget Impossibility

In conditional logit, the score is the chosen input profile minus its conditional mean, and conditional Fisher information is their covariance. Assume a profile-diameter bound D_x uniform over dates and histories. The following inequality bounds information through residual contestability.

Proposition 1 - Comparison controls information. For every date t ,

$$\text{tr} \mathcal{I}_t(\beta) \leq D_x^2 \varepsilon_t(\beta). \quad (4)$$

The conditional mean minimizes expected squared distance. Comparing it with the current leader's profile proves the bound. Summing gives a bound of $D_x^2 B(T)$ on cumulative conditional Fisher information. Residual contestability limits the information each allocation can supply; it does not guarantee informative input variation.

For two conditional recipient laws, let h_t^2 denote their one-period squared Hellinger distance:

$$h_t^2(\beta, \beta') = \sum_{i=1}^n \left(\sqrt{p_{it}(\beta)} - \sqrt{p_{it}(\beta')} \right)^2.$$

Assume local equivalence, meaning that the one-period laws have the same support after every history. Also assume comparison-dominated separation: for each β and β' there is a finite constant such that

$$h_t^2(\beta, \beta') \leq K_{\beta, \beta', \varepsilon_t}(\beta)$$

for every date and history, also after reversing the parameters. Conditional logit with uniformly bounded profile diameter satisfies this condition (Appendix A). Bounded Θ additionally gives the uniform coefficient in (6); state indices may be unbounded.

Theorem 1 - Finite comparison-budget impossibility. Suppose the design is common and predictable, local equivalence and comparison-dominated separation hold, and

$$B_\infty(\beta) < \infty \quad \mathbb{P}_\beta\text{-almost surely} \quad \text{for every } \beta \in \Theta. \quad (5)$$

1. For every $\beta, \beta' \in \Theta$, the complete single-history laws are mutually absolutely continuous on $\mathcal{F}_\infty$.
2. No sequence of estimators based on one realized history can be consistent at every β for any nonconstant contribution functional.
3. No sequence of tests can distinguish two contribution parameters with both type-I and type-II errors converging to zero.
4. For a nonconstant contribution functional, no confidence sets can have coverage tending to one and diameter tending to zero in probability at every β .

Proof. Finite comparison makes the cumulative conditional Hellinger process finite. The filtered-space Hellinger criterion gives absolute continuity of the law under β with respect to the law under β' . Reversing the parameters gives equivalence (Jacod and Shiryaev, 2003; Gabrielyan,

2011). Convergence to a constant in probability transfers between equivalent laws, so one estimator cannot converge to different contribution values. Testing and confidence sets use the same argument. Appendix B supplies the details.

Corollary 1 - Predictive merging. On the canonical allocation-history space, under Theorem 1, the conditional laws of the entire future allocation sequence given F_t merge in total variation as t grows, almost surely under both laws for each fixed parameter pair. Forecasts agree while universal consistent attribution remains impossible.

Proof. The finite-alphabet history space admits regular conditional future laws. Theorem 1 gives $P_\beta \sim P_{\beta'}$. Blackwell–Dubins (1962) gives merging under one law; equivalence transfers the null exception to the other.

Predictive merging alone permits parameter learning: equivalent prior mixtures over Bernoulli biases can merge while sample means recover bias. Distinct fixed biases induce singular path laws. Theorem 1 concerns equivalence between fixed-parameter laws. Independent replications can support consistency under Appendix E.1’s identification and regularity conditions.

For bounded Θ , conditional logit also gives a finite-horizon result:

$$D_{\text{KL}}(\mathbb{P}_\beta^T \parallel \mathbb{P}_{\beta'}^T) \leq K_{X,\theta} \|\beta - \beta'\|^2 \mathbb{E}_\beta B_T(\beta). \quad (6)$$

For differentiable scalar functionals with a nonzero directional derivative, Le Cam’s bound gives a local two-point minimax lower bound. Logarithmic expected-budget growth gives separation of order $(\log T)^{-1/2}$, under Appendix C’s conditions. This is an information limit, not an attainable or pointwise rate. The almost-sure budget condition (5) need not imply a finite expected budget.

Strong reinforcement supplies a sharp primitive case. In a reinforced market, past reward enters through a positive feedback function $g(a + N_i(t))$. If

$$\sum_{m=0}^{\infty} \frac{1}{g(a+m)} < \infty. \quad (7)$$

then an exponential-clock construction yields a finite random time after which one alternative receives every subsequent reward (Oliveira, 2009; Gottfried and Grosskinsky, 2024). The remaining probability outside the winner is summable, so B_{∞} is finite and Theorem 1 applies. For polynomial feedback with exponent ρ , this is the superlinear regime $\rho > 1$. Linear preferential attachment may generate power laws without satisfying the exact impossibility condition. The Technical Appendix gives both the exponential-clock proof and an independent common-support proof.

For fixed input profiles, latent initial position creates a separate problem. Let λ_i summarize inherited visibility, distribution access, sponsorship, or reputation. Allocation becomes

$$p_{it}(\beta, \lambda) = \frac{\exp(x_i^{\top} \beta + \lambda_i + s_{it})}{\sum_{j=1}^n \exp(x_j^{\top} \beta + \lambda_j + s_{jt})}. \quad (8)$$

For any admissible vector d , transform the parameters as

$$\beta^{(d)} = \beta + d, \quad \lambda_i^{(d)} = \lambda_i - x_i^{\top} d. \quad (9)$$

The composite index $x_i^{\top} \beta + \lambda_i$ is unchanged, so every observable allocation probability is unchanged. Contribution and position are then exactly observationally equivalent. The single-history theorem and this attribution gauge are distinct: the first is a learning impossibility even with observed position; the second is point non-identification when position is latent.

4. Shadow Futures and the Self-Reinforcing Market

A shadow future is a possible alternative finite-horizon allocation record under the same structural parameter, initial state, and predictable input and state rules, giving an agent a different

reward. Adaptive profiles may differ across records because the rules react to past allocations. These histories supply comparisons needed to estimate model-defined intervention effects. Labor and entrepreneurial markets can combine real effort, repeated entry, and skewed rewards. A creator produces a video, a contractor submits a proposal, and a founder builds a company. Each action can improve success probabilities. The allocation channel can then turn early outcomes into future exposure: recommendations train later recommendations, first customers help attract later customers, and citations produce visibility.

The winner may have contributed more. Under the theorem's conditions, one history cannot consistently recover how much more at every admissible parameter. Risk-bearing may justify prospective premiums and large prizes to induce experimentation (Kerr, Nanda, and Rhodes-Kropf, 2014). A realized jackpot need not be a sufficient statistic for causal contribution. Incentives and retrospective attribution are different questions.

This motivates a distinction between economic and epistemic monopoly. Economic monopoly concerns durable control over allocation conditions such as price, access, compatibility, or a route to market. Epistemic monopoly concerns control over the production of the comparison paths needed to explain allocation. The phrase has prior uses in knowledge governance (Meghani, 2021); here it is defined narrowly, relative to a stated contribution estimand and information set. The two can coincide, but neither implies the other. Thousands of nominal sellers may share one ranking and one distribution history. A regulated monopoly could preserve learning through randomized pilots, independent evaluation, and open experimentation. Competition has an attribution role alongside its price and discovery roles (Hayek, 2002). Separate channels can supply independent replications of a common informative allocation experiment under Appendix E.1's conditions. Portability, multihoming, interoperability, and

randomized exposure may support these conditions. Adaptive experiments designed for policy choice illustrate the role of allocation in learning (Kasy and Sautmann, 2021). Institutional plurality alone is insufficient.

The Music Lab experiments give a concrete analogue. The same songs were placed into parallel cultural markets, and stronger social influence increased inequality and cross-market unpredictability while quality remained only partially determinative (Salganik, Dodds, and Watts, 2006). The experiment does not prove the theorem for labor or entrepreneurship. It shows what a single market deletes: the terminal outcomes the same object would have occupied in other reinforced worlds.

5. Competition Policy and the Limits of Attribution

The theorem separates measurement from intervention. Audits, credentials, reputation scores, and cryptographic proofs can reduce fraud or hidden action. They do not reproduce allocations under another early sequence. Verification establishes execution; attribution requires informative comparison.

Competition policy can therefore be understood partly as evidence policy. A merger or platform design may reduce the number or independence of routes through which alternatives receive exposure and accumulate records. Two channels that look duplicative in a static cost calculation may be informational complements. Combining them can save fixed costs while eliminating a control group. The theorem does not make every merger harmful, and the welfare effect depends on decision errors and the cost of preserving independent channels. It identifies a question ordinary concentration measures can miss: how many genuinely independent histories remain available?

Public-economics models often distinguish productive returns from rent-seeking, zero-sum bargaining, or economic rents (Piketty, Saez, and Stantcheva, 2014; Rothschild and Scheuer, 2016). These models do not generally require tax authorities to recover each individual's exact causal contribution. The result here places a limit on a stronger benchmark: a tax rule that seeks to separate the contribution-attributable and position-attributable components of each realized reward. If observationally equivalent economies assign different contribution values to the same observable reward, no rule based on that record can implement the benchmark in both.

This does not invalidate progressive taxation, excess-profit taxes, or optimal-tax rules designed under uncertainty. It limits the stronger claim that the realized market record permits an exact ex post division between earned contribution and positional advantage. Tax policy must therefore rely on observable proxies, identified sets, or rules that do not require this decomposition.

6. Conclusion

A market can observe productive inputs, reward them directly, and still fail to reveal how much they contributed to the outcome. The failure arises when reward changes future exposure. As the allocation process reinforces its own history, the realized record grows while the comparisons needed for attribution disappear.

The comparison budget links residual contestability to statistical information. Under Theorem 1's conditions, a finite budget yields equivalent history laws and prevents universal consistent attribution, asymptotically perfect testing, and shrinking confidence sets with coverage tending to one everywhere. Conditional forecasts nevertheless merge: predictive agreement need not recover input effects. Strong reinforcement supplies one sufficient case; fixed inputs with latent position create a distinct failure of point identification.

Productive inputs may be public, verified, and causally effective. The missing comparisons are alternative allocation histories: shadow futures. Independent, informative routes through a market can preserve some of these comparisons, while concentration can eliminate them without concealing conventional data.

Competition policy should account for lost comparison opportunities; tax policy should avoid requiring an exact contribution-position decomposition of realized rewards. A self-reinforcing market may continue allocating reward after it has ceased producing enough evidence to explain that allocation.

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Technical Appendix

This appendix proves the information bounds, finite-budget theorem, finite-horizon lower bound, and strong-reinforcement result, and records the auxiliary replication and contribution-position arguments.

Canonical experiment and notation

All probability laws are defined on the same raw filtered measurable space $(\Omega, \mathcal{F}_\infty, (\mathcal{F}_t)_{t \geq 0})$, where $\mathcal{F}_\infty = \sigma(\cup_{t \geq 0} \mathcal{F}_t)$. The sigma-fields are not completed separately under different parameter values, and the initial law on $\mathcal{F}_0$ is common across parameters.

The verified-input profiles x_{it} and state indices s_{it} are common predictable functions of the canonical past. On a fixed realized history, their values do not change when β changes. All parameter-dependent randomness in the next observed allocation enters through the conditional recipient kernel $p_i(\beta)$. If additional observed covariates or state innovations have β -dependent conditional laws, their conditional Hellinger increments must be included in the statistical experiment.

Appendix A. Information Bounds for Conditional Logit

A.1 Fisher information

Let

$$\mu_t(\beta) = \sum_i p_{it}(\beta) x_{it}.$$

For any fixed vector a ,

$$\text{trVar}_{p_t(\beta)}(x_{J,t}) = \mathbb{E}_{p_t(\beta)} \|x_{J,t} - \mu_t(\beta)\|^2 \leq \mathbb{E}_{p_t(\beta)} \|x_{J,t} - a\|^2.$$

Choose $a = x_{w_t t}$, where w_t maximizes $p_{it}(\beta)$. Then

$$\text{tr} \mathcal{I}_t(\beta) \leq \sum_{j \neq w_t} p_{jt}(\beta) \|x_{jt} - x_{w_t t}\|^2 \leq D_X^2 \varepsilon_t(\beta),$$

which proves Proposition 1's information bound.

A.2 KL divergence

Write $\delta = \beta' - \beta$ and

$$A_t(u) = \log \sum_i \exp(x_{it}^\top u + s_{it}).$$

The conditional KL divergence is the Bregman divergence

$$D_{\text{KL}}(p_t(\beta) \| p_t(\beta')) = A_t(\beta') - A_t(\beta) - \nabla A_t(\beta)^\top \delta.$$

Taylor's theorem gives

$$D_{\text{KL}}(p_t(\beta) \| p_t(\beta')) = \int_0^1 (1-r) \delta^\top \text{Var}_{p_t(\beta+r\delta)}(x_{J,t}) \delta \, dr. \quad (\text{A1})$$

Let w_t maximize $p_{it}(\beta)$. Along the exponential tilt,

$$\frac{1 - p_{w_t t}(\beta + r\delta)}{p_{w_t t}(\beta + r\delta)} \leq \exp(D_X \|\delta\|) \frac{\varepsilon_t(\beta)}{1 - \varepsilon_t(\beta)}.$$

Because $1 - \varepsilon_t(\beta) = \max_i p_{it}(\beta) \geq 1/n$,

$$1 - \max_i p_{it}(\beta + r\delta) \leq n \exp(D_X \text{diam}(\Theta)) \varepsilon_t(\beta).$$

Applying Proposition 1 inside (A1) and integrating $1 - r$ over $[0,1]$ gives (6) with $K_{X,\Theta} =$

$\frac{1}{2} n D_X^2 \exp(D_X \text{diam} \Theta)$, for bounded Θ . For a fixed pair, replace $\text{diam} \Theta$ by $\|\beta - \beta'\|$. Since

$h_t^2 \leq D_{\text{KL}}$, comparison-dominated separation follows with the pairwise coefficient multiplied

by squared parameter distance.

Finally,

$$\log \frac{p_{J,t}(\beta')}{p_{J,t}(\beta)} = \delta^\top x_{J,t} - [A_t(\beta') - A_t(\beta)].$$

The bracketed term is constant conditional on $\mathcal{F}_t$. Its conditional variance is therefore

$$\delta^\top \mathcal{I}_t(\beta) \delta \leq \|\delta\|^2 D_X^2 \varepsilon_t(\beta),$$

which proves the auxiliary likelihood-ratio variance bound.

Appendix B. Proof Details for Theorem 1

Fix $\beta, \beta' \in \Theta$. Write $Q = \mathbb{P}_\beta$ and $P = \mathbb{P}_{\beta'}$, and let Q_T and P_T denote their restrictions to $\mathcal{F}_T$. By local equivalence, $Q_T \sim P_T$ for every finite T . Define the finite-horizon likelihood ratio

$$Z_T = \frac{dQ_T}{dP_T} = \prod_{t=0}^{T-1} \frac{p_{J_{t+1},t}(\beta)}{p_{J_{t+1},t}(\beta')} \quad (\text{B1})$$

Let

$$R_{t+1} = \frac{Z_{t+1}}{Z_t}.$$

Conditional on $\mathcal{F}_t$,

$$\mathbb{E}_P[\sqrt{R_{t+1}} \mid \mathcal{F}_t] = \sum_{i=1}^n \sqrt{p_{it}(\beta)p_{it}(\beta')} \quad (\text{B2})$$

Hence

$$2(1 - \mathbb{E}_P[\sqrt{R_{t+1}} \mid \mathcal{F}_t]) = \sum_{i=1}^n \left(\sqrt{p_{it}(\beta)} - \sqrt{p_{it}(\beta')} \right)^2 = h_t^2(\beta, \beta') \quad (\text{B3})$$

In this discrete-time, locally equivalent filtered experiment, the order-one-half Hellinger criterion states that

$$Q \ll P \quad \Leftrightarrow \quad \sum_{t=0}^{\infty} h_t^2(\beta, \beta') < \infty \quad Q\text{-almost surely} \quad (\text{B4})$$

This is the discrete-time specialization of the filtered-space criterion in Jacod and Shiryaev (2003, Chapter IV) and Gabrielyan (2011). Comparison-dominated separation and condition (5) give

$$\sum_{t=0}^{\infty} h_t^2(\beta, \beta') \leq K_{\beta, \beta'} B_{\infty}(\beta) < \infty \quad \mathbb{P}_{\beta}\text{-almost surely.}$$

Applying (B4) with $Q = \mathbb{P}_{\beta}$ and $P = \mathbb{P}_{\beta'}$ yields $\mathbb{P}_{\beta} \ll \mathbb{P}_{\beta'}$. Reversing β and β' , and using $B_{\infty}(\beta') < \infty$ $\mathbb{P}_{\beta'}$ -almost surely, gives $\mathbb{P}_{\beta'} \ll \mathbb{P}_{\beta}$. Thus

$$\mathbb{P}_{\beta} \sim \mathbb{P}_{\beta'} \quad \text{on } \mathcal{F}_{\infty} \quad (\text{B5})$$

For the estimator conclusion, let $\hat{\psi}_T$ be $\mathcal{F}_T$ -measurable and suppose $\hat{\psi}_T \rightarrow \psi(\beta)$ in $\mathbb{P}_{\beta}$ -probability. For $\epsilon > 0$, define

$$A_T(\epsilon) = \{d(\hat{\psi}_T, \psi(\beta)) > \epsilon\}.$$

Then $\mathbb{P}_{\beta}(A_T(\epsilon)) \rightarrow 0$. Since $\mathbb{P}_{\beta'} \ll \mathbb{P}_{\beta}$, absolute continuity implies $\mathbb{P}_{\beta'}(A_T(\epsilon)) \rightarrow 0$. The same estimator therefore converges to $\psi(\beta)$ under β' and cannot also be consistent for a different value $\psi(\beta')$.

For testing, let the $\mathcal{F}_T$ -measurable variable $\phi_T = 1$ denote rejection of β in favor of β' . If $\mathbb{P}_{\beta}(\phi_T = 1) \rightarrow 0$, absolute continuity implies $\mathbb{P}_{\beta'}(\phi_T = 1) \rightarrow 0$. Hence $\mathbb{P}_{\beta'}(\phi_T = 0) \rightarrow 1$, so the type-II error tends to one.

For confidence sets, choose β, β' with

$$\Delta = d(\psi(\beta), \psi(\beta')) > 0.$$

Suppose the $\mathcal{F}_T$ -measurable random sets C_T have coverage tending to one at every parameter and $\text{diam}_d(C_T) \rightarrow 0$ in probability at every parameter. Under β , define

$$E_T = \left\{ \psi(\beta) \in C_T, \text{diam}_d(C_T) < \frac{\Delta}{2} \right\}.$$

Then $\mathbb{P}_\beta(E_T) \rightarrow 1$. Equivalence implies $\mathbb{P}_{\beta'}(E_T) \rightarrow 1$. But on E_T , the set C_T cannot contain $\psi(\beta')$, contradicting coverage under β' . This proves the confidence-set conclusion.

Appendix C. Finite-Horizon Lower Bound

The chain rule for conditional KL divergence gives

$$D_{\text{KL}}(\mathbb{P}_\beta^T \parallel \mathbb{P}_{\beta'}^T) = \mathbb{E}_\beta \sum_{t=0}^{T-1} D_{\text{KL}}(p_t(\beta) \parallel p_t(\beta')).$$

Appendix A and the chain rule prove (6). Set $\Delta_\psi = |\psi(\beta) - \psi(\beta')|$ and $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$. Le Cam's two-point inequality gives

$$\max_{\vartheta \in \{\beta, \beta'\}} \mathbb{P}_\vartheta(|\hat{\psi}_T - \psi(\vartheta)| \geq \Delta_\psi/2) \geq \frac{1 - \|\mathbb{P}_\beta^T - \mathbb{P}_{\beta'}^T\|_{\text{TV}}}{2}.$$

Pinsker's inequality gives

$$\|\mathbb{P}_\beta^T - \mathbb{P}_{\beta'}^T\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D_{\text{KL}}(\mathbb{P}_\beta^T \parallel \mathbb{P}_{\beta'}^T)},$$

which yields the two-point lower bound stated in Section 3 of the main article.

For the local bound, fix an interior β_0 , put $b_T = \mathbb{E}_{\beta_0} B_T(\beta_0) \rightarrow \infty$, and choose a unit direction v with $\nabla \psi(\beta_0)^\top v \neq 0$. Set

$$\beta_T = \beta_0 + \frac{hv}{\sqrt{b_T}}$$

with sufficiently small fixed $h > 0$. Differentiability at β_0 gives

$$|\psi(\beta_T) - \psi(\beta_0)| = \frac{|h \nabla \psi(\beta_0)^\top v|}{\sqrt{b_T}} + o(b_T^{-1/2}).$$

Equation (6) bounds KL by $K_{X,\theta} h^2$. Choosing h small keeps total variation below one. The two-point inequality bounds maximal error probability away from zero at separation proportional to $b_T^{-1/2}$ over $\{\beta_0, \beta_T\}$. This local minimax bound does not prohibit pointwise superefficiency.

Appendix D. Strong Reinforcement

Consider a finite set of agents with fixed verified-input profiles x_i . Let

$$R_{i,t+1} = \mathbf{1}\{J_{t+1} = i\}, \quad N_i(t) = \sum_{r=1}^t R_{i,r}, \quad N_i(0) = 0. \quad (D1)$$

Let $a > 0$ and let g be positive and nondecreasing. Conditional allocation is

$$p_i(\beta) = \frac{\exp(x_i^\top \beta) g(a + N_i(t))}{\sum_{j=1}^n \exp(x_j^\top \beta) g(a + N_j(t))}. \quad (D2)$$

Assume strong reinforcement:

$$\sum_{m=0}^{\infty} \frac{1}{g(a + m)} < \infty. \quad (D3)$$

Write

$$q_i = \exp(x_i^\top \beta) > 0.$$

For each agent i and occupancy level $m \geq 0$, let $E_{i,m}$ be independent exponential random variables with rate $q_i g(a + m)$, and define

$$T_i = \sum_{m=0}^{\infty} E_{i,m}.$$

Condition (D3) gives

$$\mathbb{E}[T_i] = \sum_{m=0}^{\infty} \frac{1}{q_i g(a + m)} < \infty,$$

so $T_i < \infty$ almost surely. The variables T_i are independent and continuously distributed, so there is almost surely a unique minimizer I^* .

The exponential-clock construction has the same embedded recipient sequence as equation (D2). Agent I^* experiences infinitely many clock rings by time T_{I^*} . Every other agent has only finitely many rings before that time. Hence all but finitely many discrete rewards go to I^* .

Let n_j denote the final count of loser j and let the winner's post-absorption count grow as m .

Then

$$\varepsilon_t(\beta) \leq \frac{\sum_{j \neq I^*} q_j g(a + n_j)}{q_{I^*} g(a + m)}.$$

Summing over m and using condition (D3) gives $B_\infty(\beta) < \infty$ almost surely.

Direct common-support proof for strong reinforcement. Let $\mathcal{M}$ be the countable set of eventually constant recipient histories. The absorption argument above gives $\mathbb{P}_\beta(\mathcal{M}) = 1$ for every finite β . Fix $\omega \in \mathcal{M}$. Suppose that after a finite prefix ending at date T , the history assigns every later reward to agent i , and let $n_j = N_j(T)$.

The finite prefix has strictly positive probability under every finite β . Conditional on that prefix, the probability that agent i receives every later reward is

$$\prod_{m=0}^{\infty} \frac{q_i g(a + n_i + m)}{q_i g(a + n_i + m) + \sum_{j \neq i} q_j g(a + n_j)}.$$

Let

$$C_{\beta, \omega} = \sum_{j \neq i} q_j g(a + n_j).$$

The tail probability becomes

$$\prod_{m=0}^{\infty} \left[1 + \frac{C_{\beta, \omega}}{q_i g(a + n_i + m)} \right]^{-1}.$$

Condition (D3) implies

$$\sum_{m=0}^{\infty} \frac{C_{\beta,\omega}}{q_i g(a + n_i + m)} < \infty.$$

The infinite-product criterion therefore makes the tail probability strictly positive. Hence $\mathbb{P}_{\beta}(\{\omega\}) > 0$ for every $\omega \in \mathcal{M}$ and every finite β . Since every $\mathbb{P}_{\beta}$ assigns probability one to the same countable set and positive probability to every atom in that set, the strong-reinforcement laws have the same positive atomic support and are mutually absolutely continuous. This independently proves the strong-reinforcement result and its estimator and testing consequences stated in the main article.

Appendix E. Auxiliary Results

E.1 Independent replication

Observe M independent copies of the β -market through fixed horizon L . Assume Θ is compact, finite-history probabilities are continuous and strictly positive in β , and the history law identifies β . Finitely many outcomes give uniform convergence of log likelihoods; Gibbs' inequality gives a unique population maximizer. Maximum likelihood is consistent as $M \rightarrow \infty$.

E.2 Contribution-position invariance

For fixed profiles and admissible transformed parameters, replacing β by $\beta + d$ and λ_i by $\lambda_i - x_i^T d$ leaves $x_i^T \beta + \lambda_i$ unchanged. Hence every allocation probability and history law is unchanged. Time-varying within-agent input variation can break this gauge because one fixed λ_i need not absorb all $x_{it}^T d$.